Fast Fourier Transform (FFT)

Interpretation

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How to Interpret DFT/FFT result?

- **Recall DFT:** $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$
- $X[k]$ **is complex valued, can compute and plot spectrum:** $|X[k]|$ **or** $|X[k]|^2$
- **Given sampling frequency** f_s**, what analog frequency does** $X[k], k = 0, 1, \dots, N-1$ **represent?**

Recall: Frequency Resolution of DFT

- **DFT:** $X[k] = X\left(e^{j\omega}\right)\Big|_{\omega_k = k\frac{2\pi}{N}}$
- **Digital frequency resolution in radians:** $\Delta\omega = \frac{2\pi}{N}$
- **Analog frequency resolution in radians/second:** $\Delta\Omega = \frac{2\pi}{N} f_s$
- **Analog frequency resolution in Hz:** $\Delta f = \frac{f_s}{N}$

DFT Interpretation

- $X[k], k = 0$ **represents DC frequency component, i.e.,** $f_0 = 0.$
- $X[k], k = 1, \dots, \frac{N}{2} - 1$ **represent positive frequencies, i.e.,**
  - $f_k = \frac{kf_s}{N}, k = 1, \dots, \frac{N}{2} - 1.$
  - $f_1 = \frac{f_s}{N}, f_2 = \frac{2f_s}{N}, \dots, f_{\frac{N}{2}-1} = \frac{f_s}{2} - \frac{f_s}{N}.$
- $X[k], k = \frac{N}{2}$ **represents the Nyquist frequency, i.e.,** $f_{\frac{N}{2}} = f_s/2.$
- $X[k], k = \frac{N}{2} + 1, \dots, N - 1$ **represent negative frequencies, i.e.,**
  - $f_k = \frac{kf_s}{N} - f_s, k = \frac{N}{2} + 1, \dots, N - 1.$
  - $f_{\frac{N}{2}+1} = -\frac{f_s}{2} + \frac{f_s}{N}, \dots, f_{N-2} = -\frac{2f_s}{N}, f_{N-1} = -\frac{f_s}{N}.$

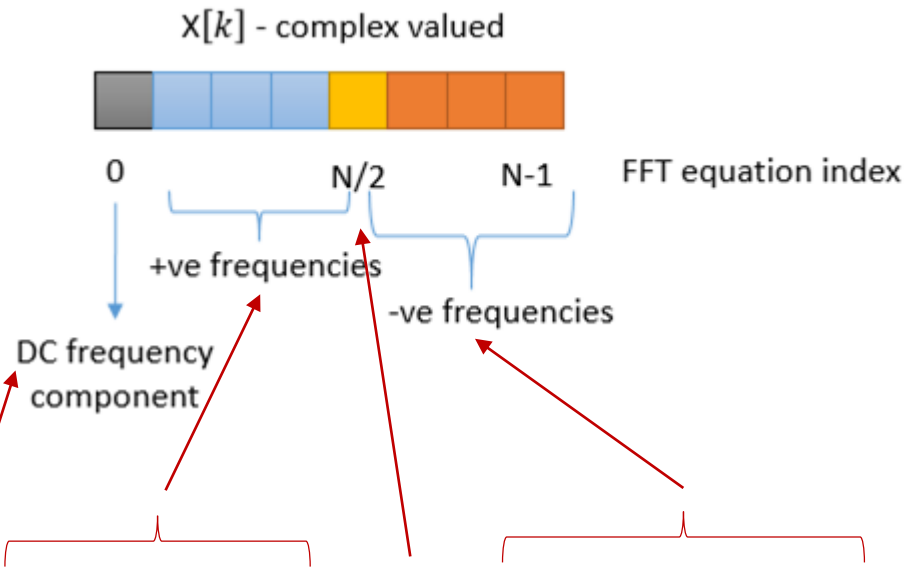
Example

- **Given the sinusoid $x(t) = sin(2\pi f t)$ with *f* = 256 Hz and sampling frequency f_s = 8192 Hz, for a *N* = 64-point DFT X[k], which X[k] has non-zero values?**

Answer

- **Given f = 256 Hz, f_s = 8192 Hz, N = 64**
- $\Delta f = \frac{f_s}{N} = \frac{8192}{64} = 128Hz$
- $X[k], k = 0$ **represents DC frequency component, i.e.,** $f_0 = 0Hz$
- $X[k], k = 1, \dots, \frac{N}{2} - 1$ **represent positive frequencies, i.e.,**
  - $f_k = \frac{kf_s}{N} = \frac{8192k}{64} = 128k = 128, 256, \dots, 3968Hz$
- $X[k], k = \frac{N}{2}$ **represents the Nyquist frequency, i.e.,**
  - $f_{\frac{N}{2}} = f_s/2 = 4096Hz$
- $X[k], k = \frac{N}{2} + 1, \dots, N - 1$ **represent negative frequencies, i.e.,**
  - $f_k = \frac{kf_s}{N} - f_s = -3968, \dots, -256, -128Hz$

Answer (cont'd)



f_k = [0, 128, 256, …, 3968, 4096, -3968, …., -256, -128] Hz

https://www.gaussianwaves.com/2015/11/interpreting-fft-results-complex-dft-frequency-bins-and-fftshift/

Answer (cont'd)

- **Recall:** $x[n] = sin\omega_0 n \overset{F}{\leftrightarrow} X\left(e^{j\omega}\right) = \frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$
- $f_k = 128k = 256Hz \Rightarrow k = 2$
- $f_k = \frac{kf_s}{N} - f_s = \left(\frac{k}{64} - 1\right)8192 = -256Hz \Rightarrow k = N - 2 = 62$
- **Therefore, *X[2]* and *X[N-2]* = *X[62]* has non-zero values.**

Question

Given the sinusoid $x(t) = \sin(2\pi ft)$ and sampling frequency $f_s = 8192$ Hz, for a $N = 128$-point DFT $X[k]$, if $X[4]$ and $X[124]$ have non-zero values, what is the signal's frequency f in Hz?

a) 128

b) 256

c) 7872

d) 7936

DFT of Real-Valued Signals

- $x[n]$ **is real-valued.**
- **DFT:** $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \ldots, N-1$
- $X[0] = \sum_{n=0}^{N-1} x[n]$ **(a real number)**
- $X\left[\frac{N}{2}\right] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi N}{N\ 2}n} = \sum_{n=0}^{N-1} x[n] e^{-j\pi n}$ **(a real number)**
- **DFT of real valued signals:** $X[k] = X^*[(-k)_N]$
  - Only $k = 1, \ldots, \frac{N}{2} - 1$ (corresponding to the positive frequencies) are needed.
  - Can derive $X[N-1] = X^*[1]$, $X[N-2] = X^*[2]$, …
- **A size *N* array can be used to store the *N*-point FFT** $X[k]$ **(real and imaginary parts) of a real-valued signal.**

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